



Genie Cost Optimization Best Practices

March 2026



**What if you could
talk with your data?**



Genie

Talk with your data

Text2Sql Agent

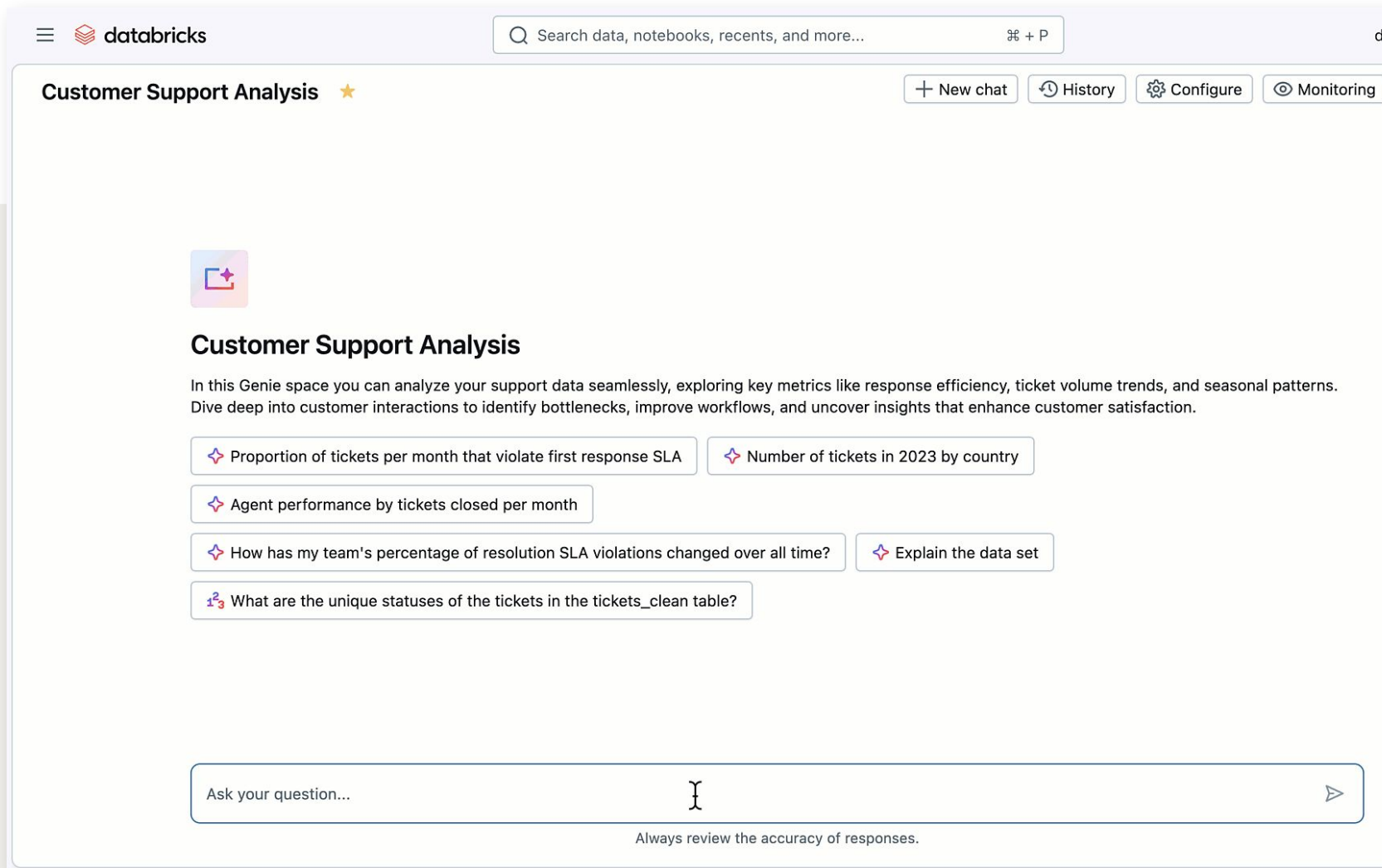
Conversational analytics
tailored to your business

Governed and secure

Leverages all data assets
on Unity Catalog through
DBSQL

Deep Reasoning

Explore trends and
validate hypothesis



The screenshot shows the Databricks Genie interface for a workspace named "Customer Support Analysis". At the top, there's a search bar and navigation links for "New chat", "History", "Configure", and "Monitoring". Below the header, a small icon represents the Genie agent. The main section is titled "Customer Support Analysis" and contains a descriptive paragraph about analyzing support data. Below this, there are five interactive buttons, each with a star icon and a question: "Proportion of tickets per month that violate first response SLA", "Number of tickets in 2023 by country", "Agent performance by tickets closed per month", "How has my team's percentage of resolution SLA violations changed over all time?", and "Explain the data set". At the bottom, there's a text input field with the placeholder "Ask your question..." and a "Send" button. A disclaimer at the very bottom states "Always review the accuracy of responses."

databricks

Search data, notebooks, recents, and more...

Customer Support Analysis ★

+ New chat History Configure Monitoring



Customer Support Analysis

In this Genie space you can analyze your support data seamlessly, exploring key metrics like response efficiency, ticket volume trends, and seasonal patterns. Dive deep into customer interactions to identify bottlenecks, improve workflows, and uncover insights that enhance customer satisfaction.

✦ Proportion of tickets per month that violate first response SLA

✦ Number of tickets in 2023 by country

✦ Agent performance by tickets closed per month

✦ How has my team's percentage of resolution SLA violations changed over all time?

✦ Explain the data set

✦ What are the unique statuses of the tickets in the tickets_clean table?

Ask your question...

Always review the accuracy of responses.



Genie Deep Reasoning

What would you like to know?

Ask your question...



Always review the accuracy of responses.

Genie forms various hypothesis and investigates each one in turn





Why was there a spike in encounters in March 2021?

Researched for 63 seconds

The spike in encounters in March 2021, reaching 453 encounters, appears to be part of a larger upward trend observed throughout 2020 and early 2021¹. This significant increase was primarily driven by a high number of outpatient and ambulatory encounters, which accounted for 282 and 112 encounters respectively².

The detailed breakdown of encounter reasons in March 2021 shows that chronic conditions like congestive heart failure and hyperlipidemia were among the most common reasons, each with 16 encounters³. However, these reasons alone do not fully explain the spike, as they represent a small portion of the total encounters.

The total claim cost for March 2021 was relatively low at 627,152.36, indicating that the spike was not due to more severe or resource-intensive cases⁴. Additionally, demographic data from previous years does not provide insights into the specific spike in March 2021⁵.

Overall, the spike in March 2021 seems to be a continuation of the increasing trend in patient encounters, with a notable contribution from outpatient and ambulatory services, but the exact underlying causes remain unclear without further context.

Research results

1. Monthly Encounter Trends
2. Encounter Classifications in March 2021
3. Encounter Reasons in March 2021
4. Monthly Encounter Cost Trends
5. Patient Demographics in March 2021

Ask Genie another research question...

Users can explore Genie's research
for each hypothesis it forms

Genie =
Text2Sql Agent
+ DBSQL



Genie =
Text2Sql Agent → Metadata is key
+ DBSQL → Query
performance is key





How does query performance look like



Databricks SQL

World-class price/performance

Up to 12x faster than alternatives
at high concurrency

Lower costs

Predictive optimizations deliver
automatic, AI-driven savings

Quicker answers

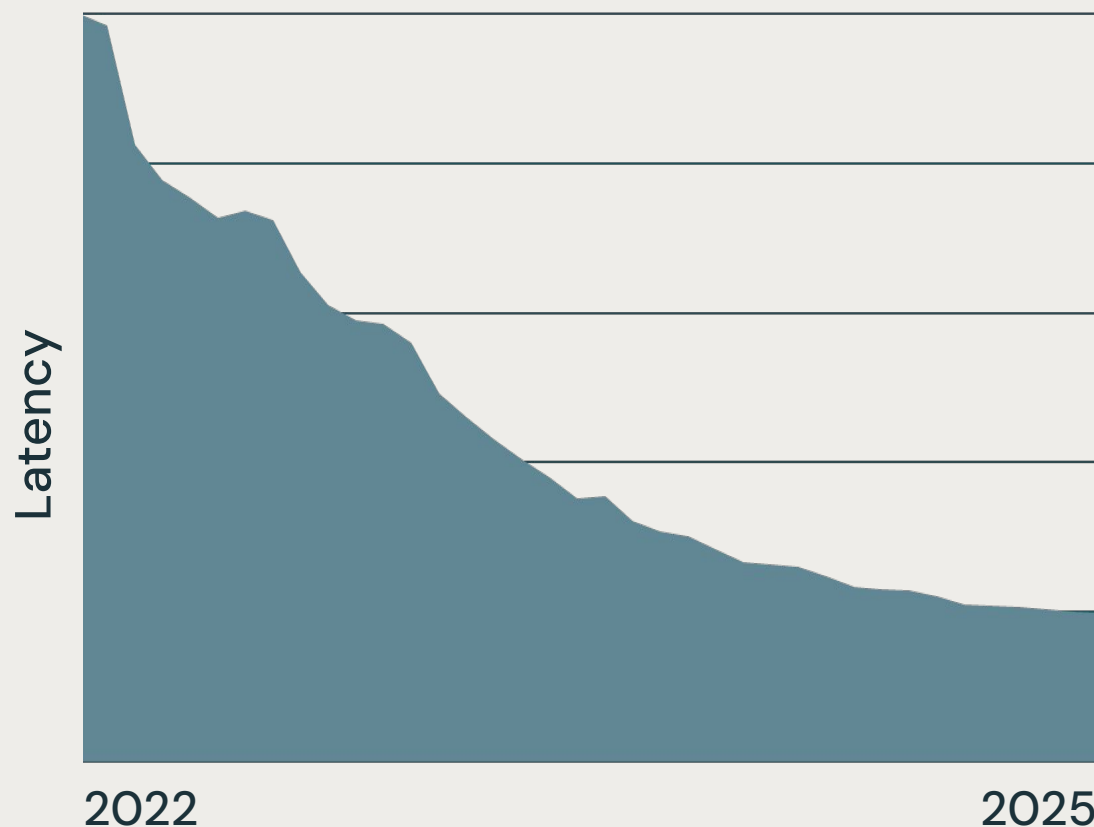
Natural-language queries that
understand your semantics

Adopted by thousands of global enterprises

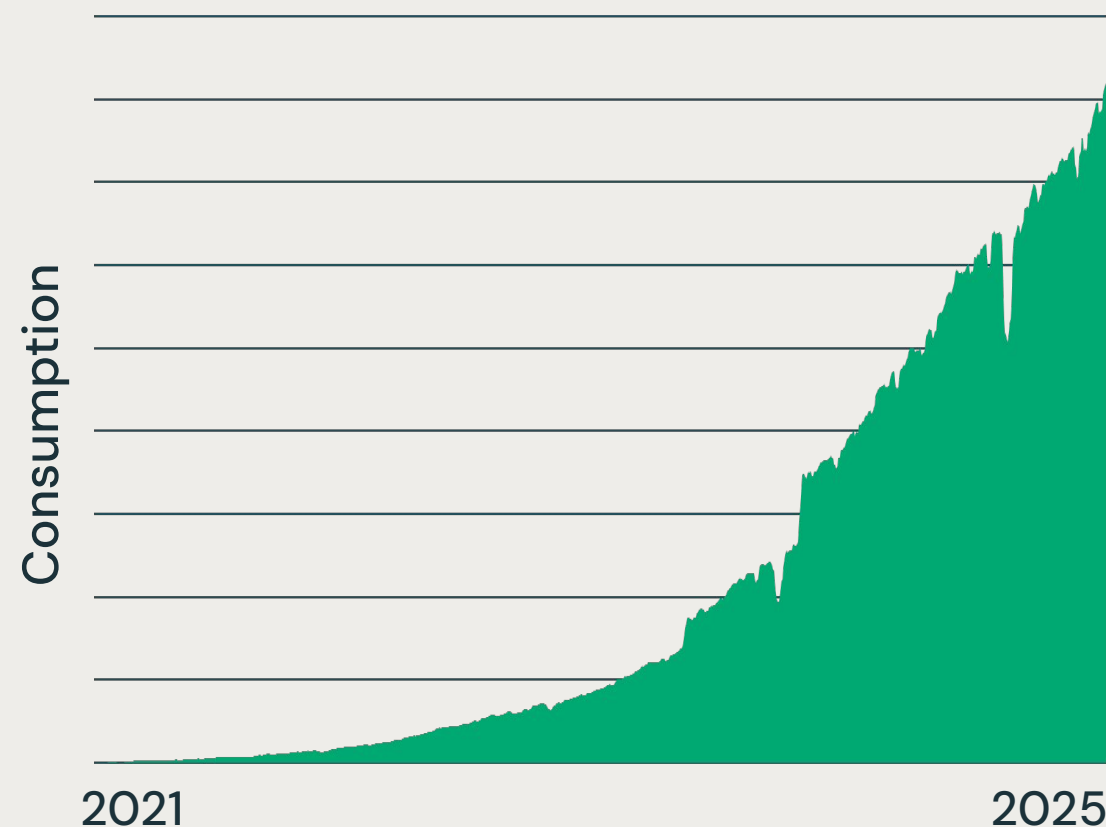


Efficient performance is driving adoption

5X faster query performance since launch



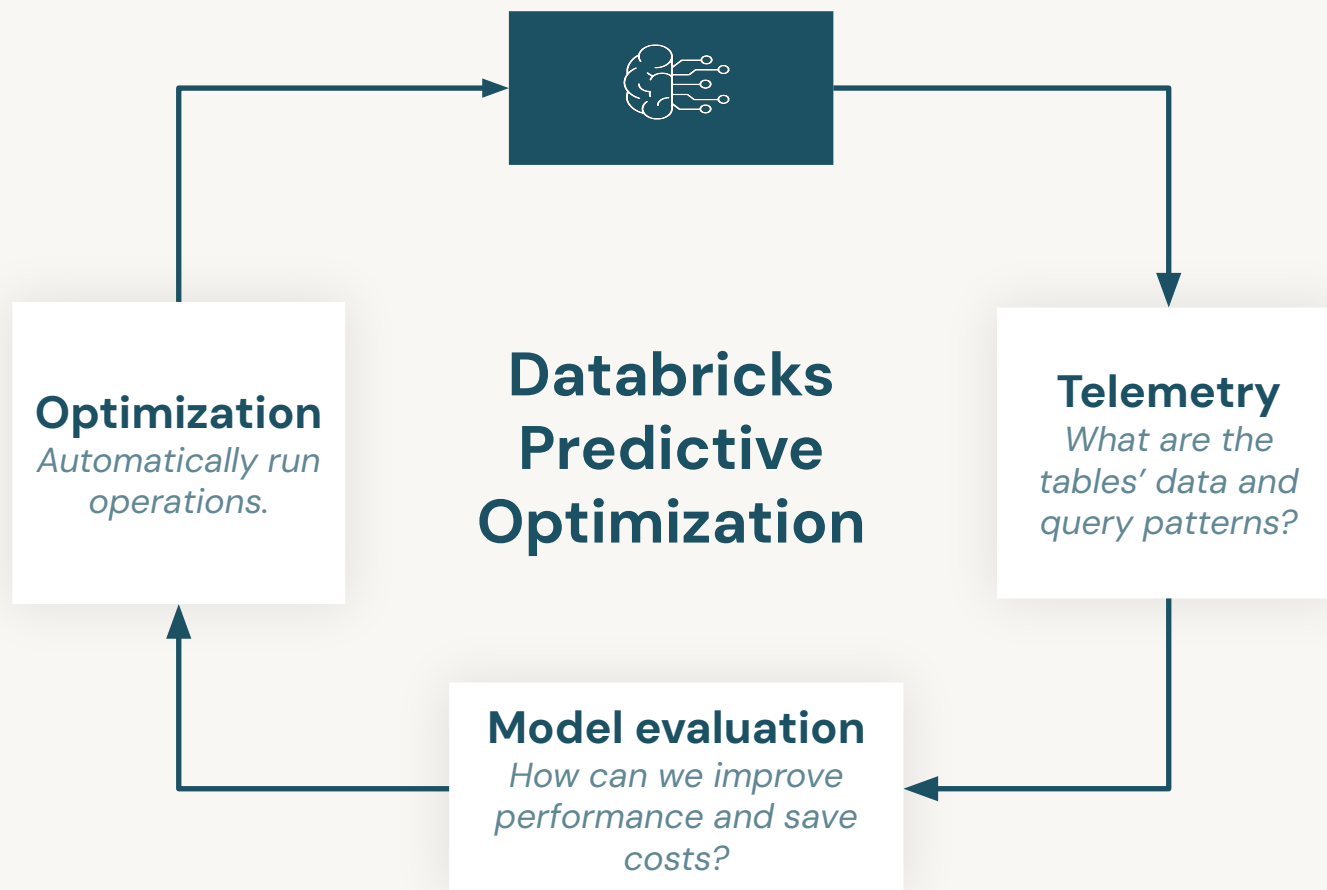
2X more consumption in the past year



AI-Optimized Data Layout Management

With Predictive Optimization

General Availability



Supported optimizations:

OPTIMIZE

VACUUM

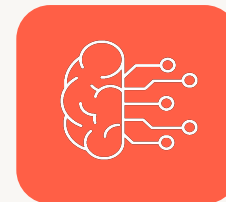
Liquid Clustering



Automatic Liquid Clustering Key Selection

With Predictive Optimization

General Availability



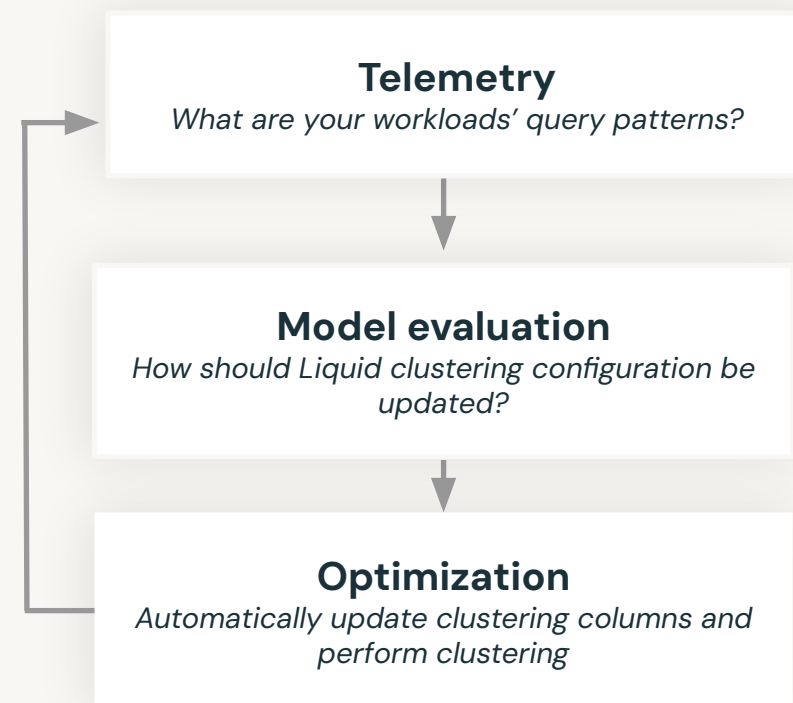
Liquid clustering
powered by the
Data Intelligence
Engine

```
> CREATE TABLE tbl CLUSTER BY AUTO
```

Fully automated:

- Clustering key selection
- Clustering on write
- Background clustering

[Detail on how it works](#)



Types of Compute in Databricks

Classic Compute vs Serverless

Classic Compute

- **Infrastructure Management Required**
 - Version Control and Upgrades
 - Infrastructure team required
- **Limited Scalability**
 - Horizontal to a degree
 - Challenging to scale down
- **Innovation bottleneck**
 - Hard to make improvements without testing
 - “Technical debt” on DBX Innovation

Use for R, Scala*, predictable workloads

Serverless Compute

- **Goodbye Infrastructure Management**
 - Serverless = “Versionless”
 - Fully managed infrastructure
- **Best scalability available**
 - Horizontal and vertical
 - Scale-to-zero capabilities!
- **Future readiness**
 - Constantly optimized
 - Enables additional capabilities (Enzyme, Photon and many more)

Use for Python, Pyspark, SQL, Java*





Choose your business goal

In classic, you choose
perf/cost via infra settings



This is hard!

Serverless is simple

Compute*

Serverless

- Choose a T-Shirt Size
- Instant Startup time!



We continuously optimize



Enriching Genie with the correct metadata

How to supercharge Genie's understanding of business semantics



Metadata Magic

The building block for everything

Harness intelligence to describe and work with your most important data

Leverage the power of all that context to make your team more efficient

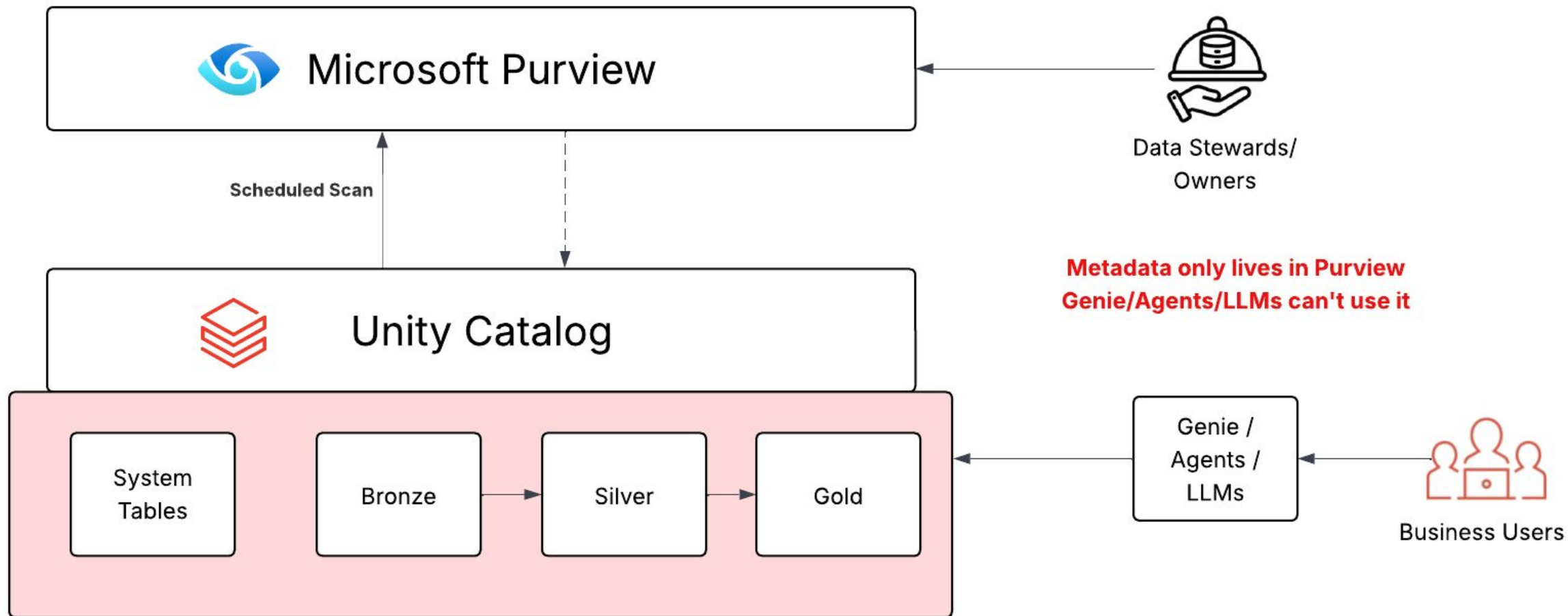
The screenshot displays the Databricks Assistant interface. At the top, a window titled "Edit the description for anthony-hongdb.test_schema.customers_100" shows a text input field with the following description: "The 'customers_100' table contains information about 100 customers, including their personal details, contact information, and subscription details. This data can be used to manage customer relationships, segment customers based on demographics, and analyze". Below the input field are buttons for "AI generate", "Preview", "Shorten text", "Lengthen text", and "Revise tone".

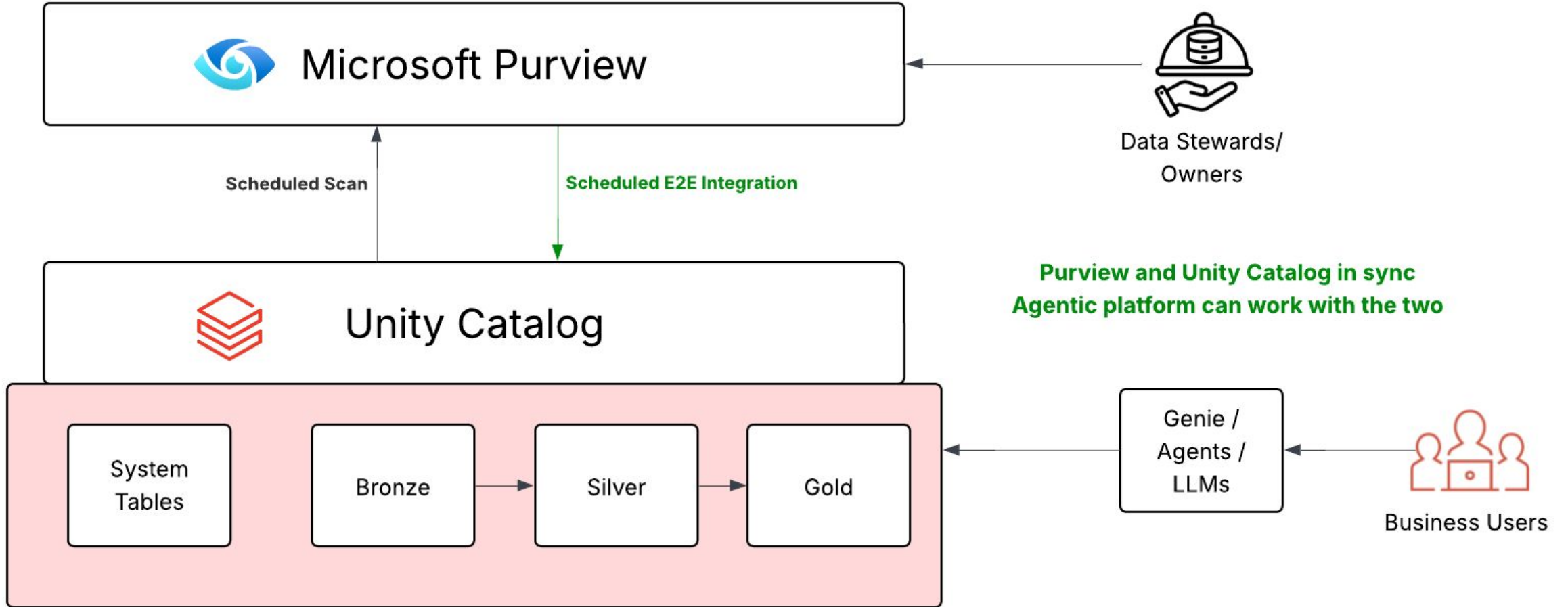
The main Assistant window is titled "Assistant" and features a blue icon with a magnifying glass and a plus sign. Below the icon, the text reads: "Databricks Assistant. Accelerate your work by diagnosing errors, suggesting code or queries, and answering questions. Check out [some examples](#) to get started. Make sure to verify any generated suggestions and [share feedback](#) so we can learn and improve." At the bottom of the Assistant window are two buttons: "Find tables to query" and "Find some queries". Below these buttons is a text input field with the placeholder "Ask assistant or type '/' for commands" and a right-pointing arrow button.

On the right side, a chat window titled "Assistant" shows a conversation. The user's message is "find tables about food inspections". The Assistant's response is: "In the provided information, there is one table that directly relates to food inspections:

- Table name: `richardt_demos.chicago_data.food_inspections`

 The 'food_inspections' table contains records of food inspections conducted in Chicago. It includes details such as the names of the DBAs and AKAs involved, their licenses, type of facilities inspected (e.g., restaurants, coffee shops, bars, etc.), and any identified risks or violations. The data also includes geographical information in the Midwest such as latitude and longitude along with the address and city where the inspection took place. If you need further information from this table or wish to perform any specific queries, please let me know what information you're interested in, and I can assist with formulating the appropriate SQL query." At the bottom of the chat window is a text input field with the placeholder "Ask assistant or type '/' for commands" and a right-pointing arrow button.





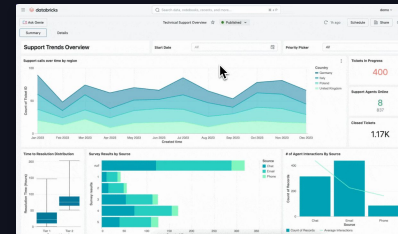
Rich Semantics

Define and share your metrics

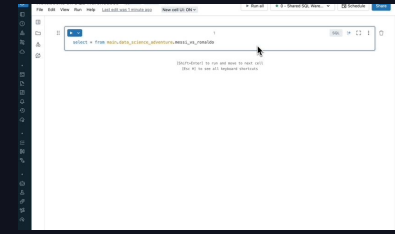
Empower all users and AI agents with unified, reusable building blocks

Reduce redundancy and establish trust with centralized governance

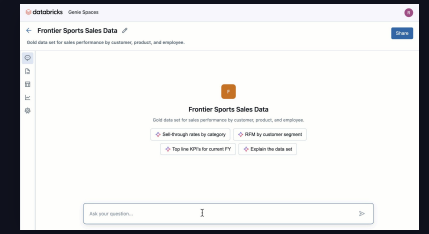
Access via REST or SQL, with native integrations for external tools in active development



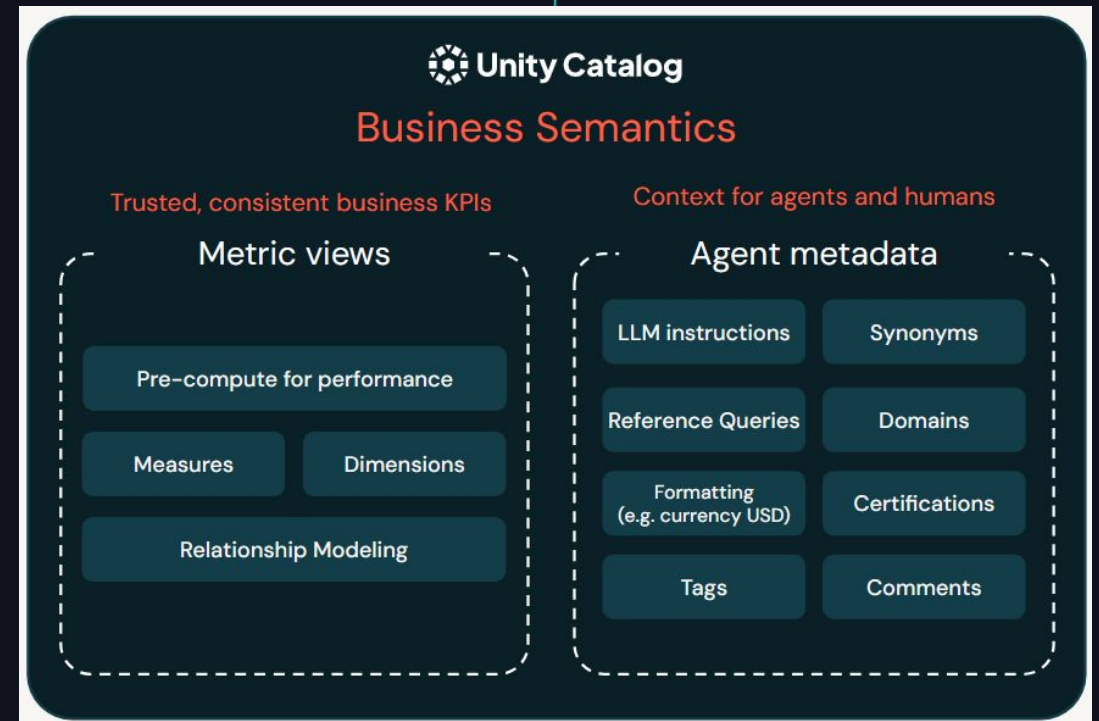
AI/BI Dashboards



Queries & Notebooks



AI/BI Genie



Genie Room =
Table Domain Expertise



What if I have multiple Genie Rooms?

Leverate Orchestration capabilities on Databricks

Agent Supervisor: The managed approach

Supervisor Agent

Design an AI system with Genie, agents, tools.

Tool-calling Agent orchestration

Build

Agent

Type

Genie Space

Agent Endpoint

Unity Catalog Function

External MCP Server

MCP and Tools for LLMs: DIY with full customization

Claude Sonnet 4.6

Tools (1)

Get code

MCP Servers

Genie Space: EDP Client Solutions Silver Data

+ Add tool

You can add up to 20 tools

system.ai
Learn more

Select an endpoint

Search

Model type

Name	Type
GPT OSS 120B databricks-gpt-oss-120b	AI Gateway
✓ Claude Sonnet 4.6 databricks-claude-sonnet-4-6	AI Gateway
Claude Sonnet 4.5 databricks-claude-sonnet-4-5	AI Gateway
Gemini 3 Flash databricks-gemini-3-flash	AI Gateway
Claude Haiku 4.5 databricks-claude-haiku-4-5	AI Gateway

Databricks One: Consumer only experience

I want to know about EDP ClientSolutions Silver: how are my c...

I want to know about EDP ClientSolutions Silver: how are my customers purchase going? Which are the top spenders?

Thought process

I will first check for relevant Genie spaces for "EDP ClientSolutions Silver" and customer purchase analysis, as this is the most cost-efficient way to answer your question. If a suitable Genie space exists, I'll use it to analyze customer purchases and identify top spenders

Found 10 results for "EDP ClientSolutions Silver, customer purchases, top spenders"

Asked: EDP Client Solutions Silver Data

EDP ClientSolutions Silver: Customer Purchase Overview and Top Spenders

Your customers' purchases are progressing well, with clear leaders in total spend. The top 10 spenders are identified below, showing both their names and resolved customer IDs for easy reference.

Top Spenders:

- Rodriguez, Figueroa and Sanchez (CUST-000001) leads with €3,580,809.81 in purchases.
- Doyle Ltd (CUST-000002) follows with €1,547,467.48.
- McClain, Miller and Henderson (CUST-000003) and Davis and Sons (CUST-000004) are also significant contributors.

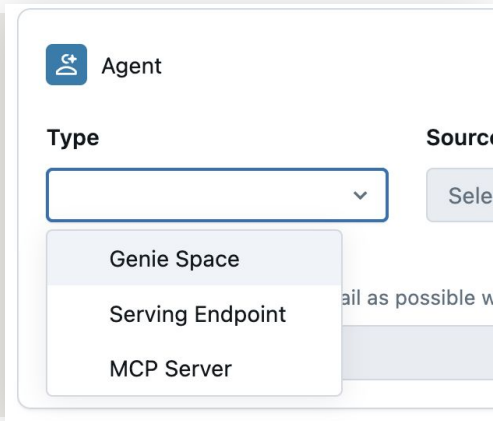
The full ranking and spend amounts are visualized below for actionable insights.

Visualization

Top 10 Customers by Total Spend (EDP ClientSolutions Silver)

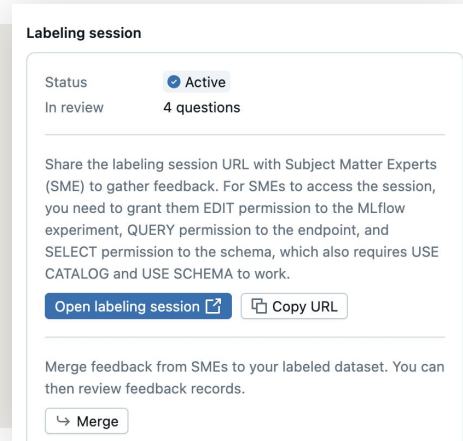


Agent Supervisor is the managed approach



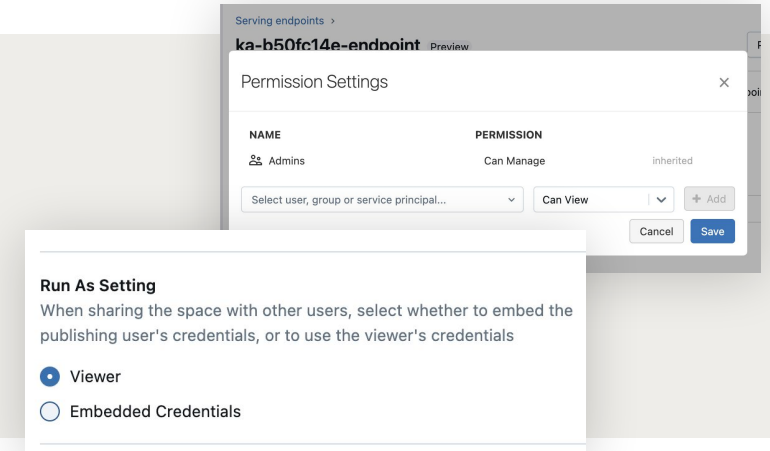
Flexible for your use case

Build agents that can reason over all your data – structured or unstructured



Agent Learning & evaluation built-in

Gather feedback from SMEs & improve the model over time



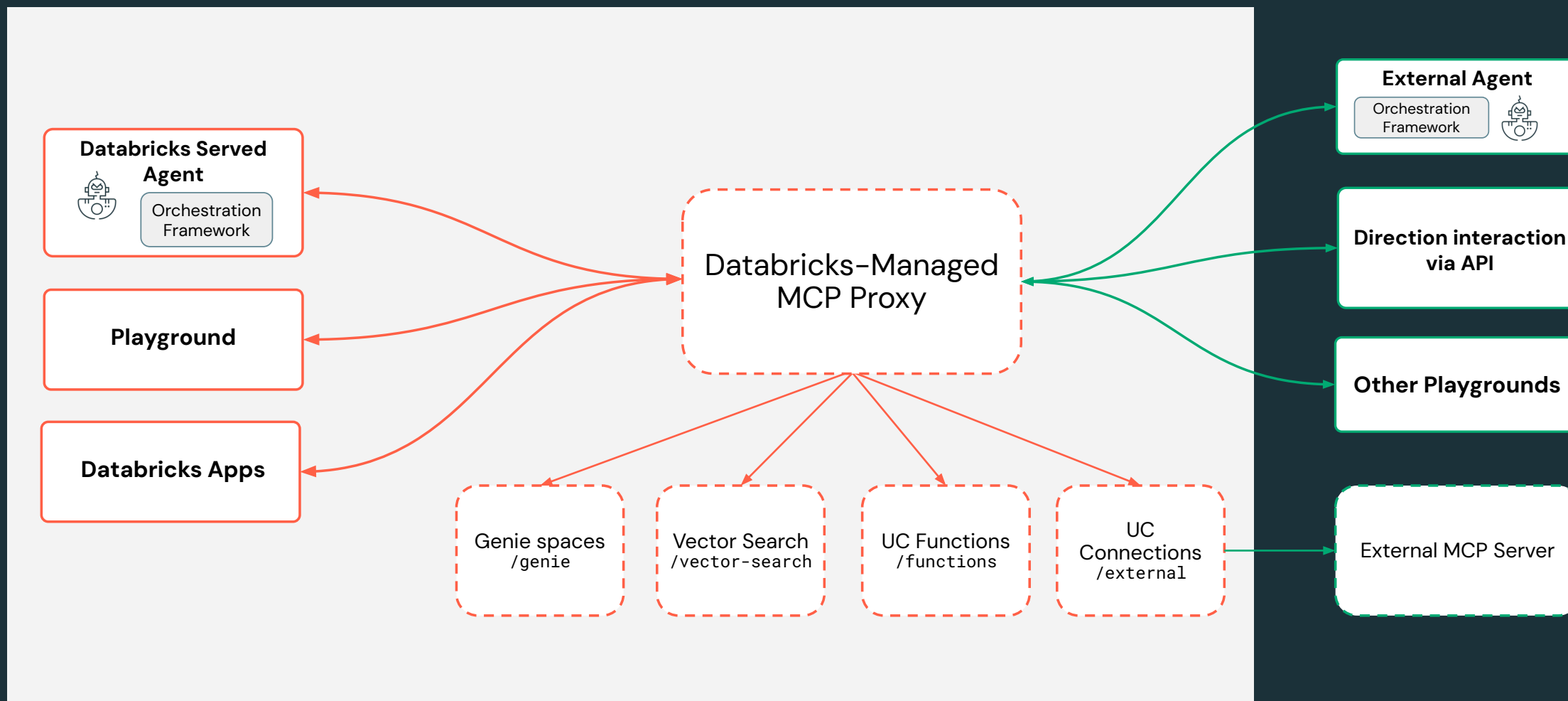
Governance & access built-in

Leverage existing permissions with each agent so you only expose data they have access to

GA on multiple Azure Regions on March 3rd



Interacting with Managed Servers



Databricks One

Data Intelligence for business teams

Intuitive consumer experience for Dashboards, Genie spaces, and Apps

Simple for administrators to quickly onboard new users

Built in access control, auditability, and governance through Unity Catalog

databricks

Databricks One D

What would you like to know?

Search for Dashboards and Genie spaces...



Search



Ask



For you

Dashboards

Genie spaces

Apps

Dashboard

Customer Support Team Revi...



Dashboard

Sales Pipeline Review



Dashboard

Portfolio Assistant



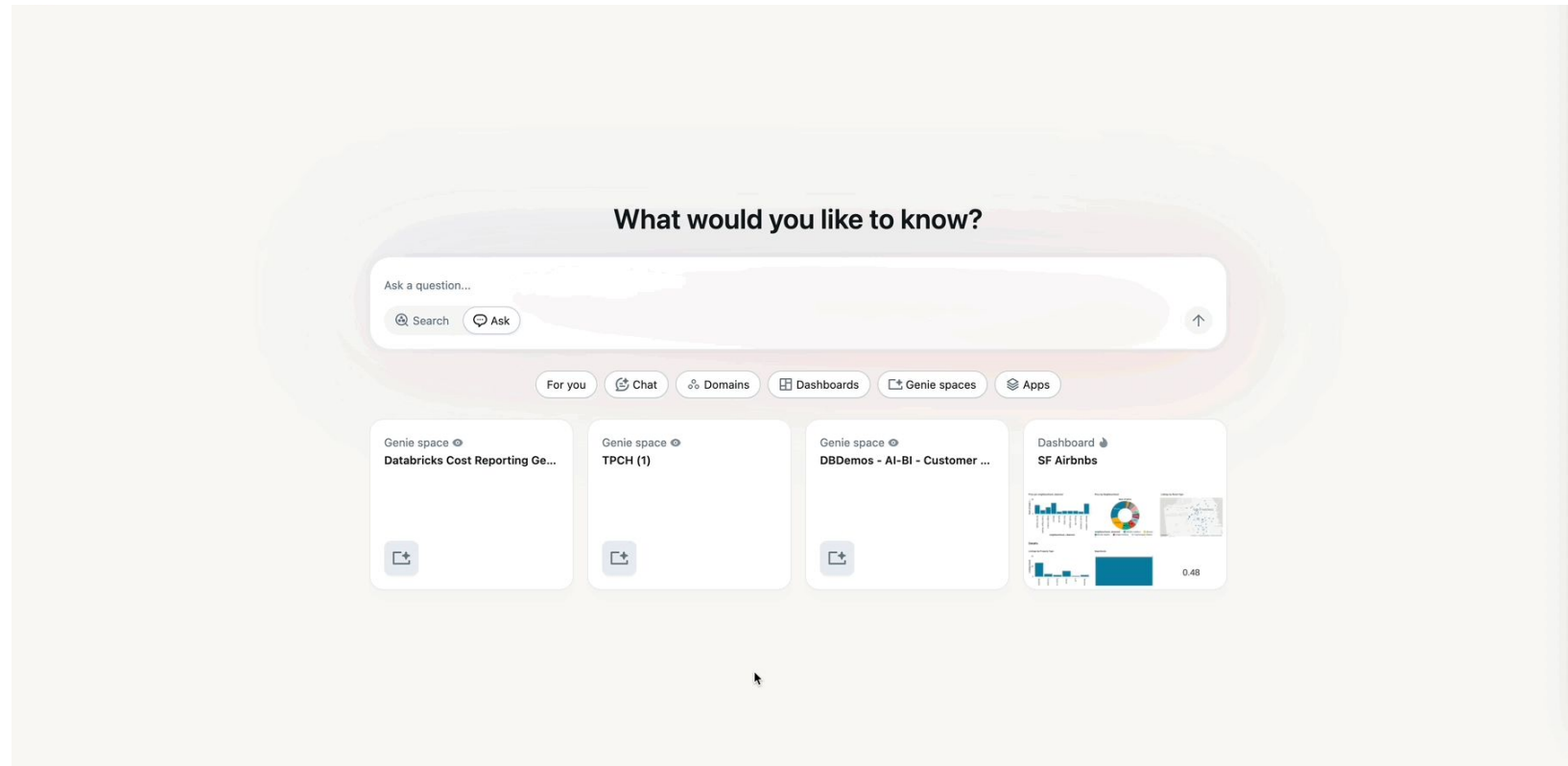
Genie space

Customer Support Review



Unified Chat (OneChat)

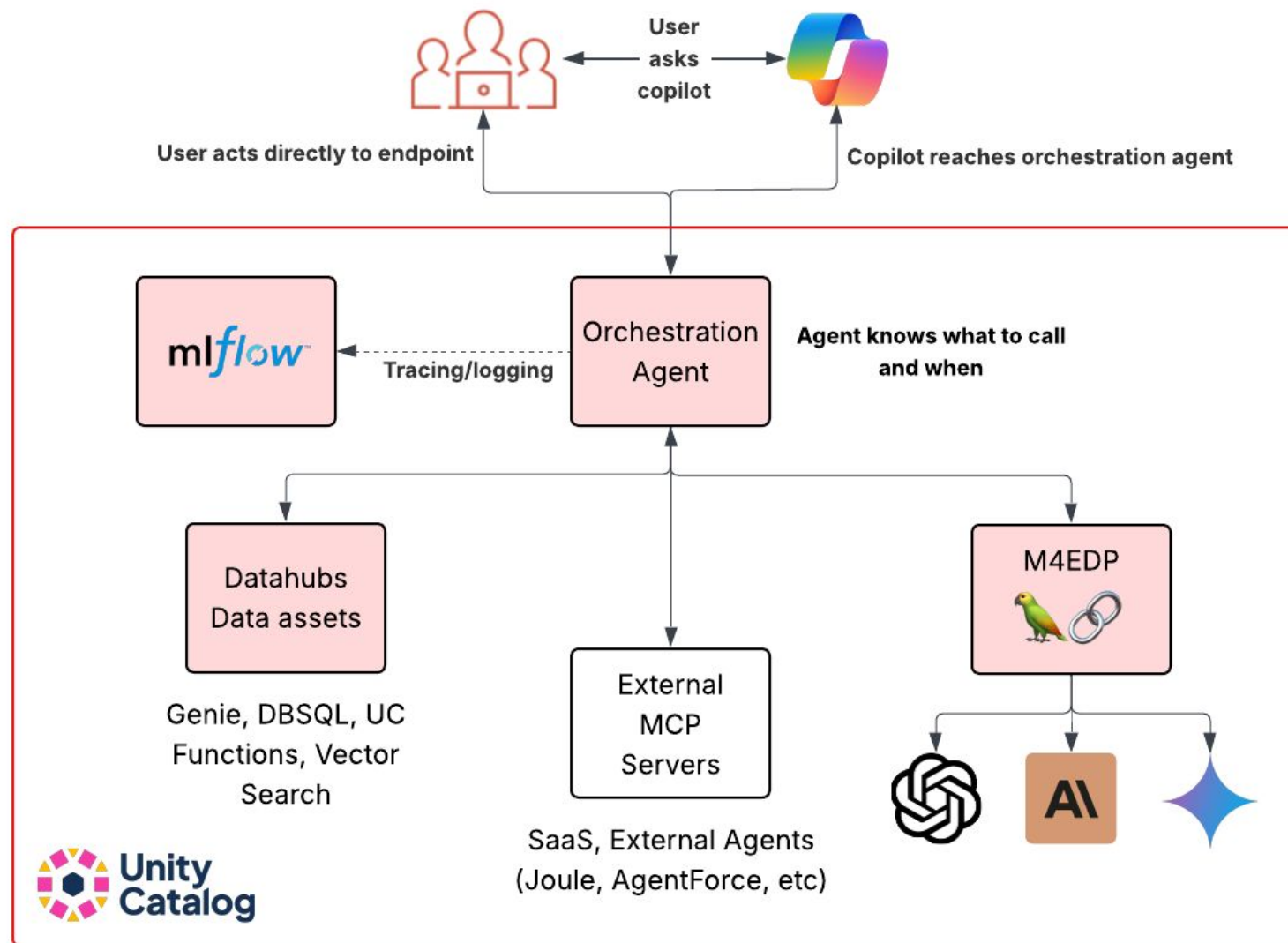
Unified Chat Experience on Databricks One with basic Genie Routing



If no pre-existing Genie space is available to the user, the system will attempt to answer the question using a version of the Data Science Agent.



An agentic orchestration vision for EDP



Genie Performance Best Practices

Combination of Efficient Queries + Efficient AI Engine

Query Optimization

- **Use SQL Serverless Compute**
 - Best User Experience (minimal turn on time)
 - Intelligent Scaling for short powerful queries
 - Cache capabilities with query profile
- **Enable Predictive Optimization**
 - Automated optimization for your Data Layer
- **Leverage Automatic Liquid Clustering**
 - Best for unpredictable query patterns, considering Genie open ended questions

AI Engine Optimization

- **UC Objects with powerful metadata**
 - Purview > UC Pilot
 - Semantic layers like UC Metric Views
- **Genie for domain expertise**
 - Don't create one genie room for everything, separate on "subdomains"
 - Edit a genie room to become an expert on its domain, not on its tables
 - Orchestrate multiple Genie rooms with **Supervisor Agent, OneChat, Tools**
- **Evaluate your Genie room with Benchmarks**
 - Have a set of business approved questions that can be evaluated for performance over time

